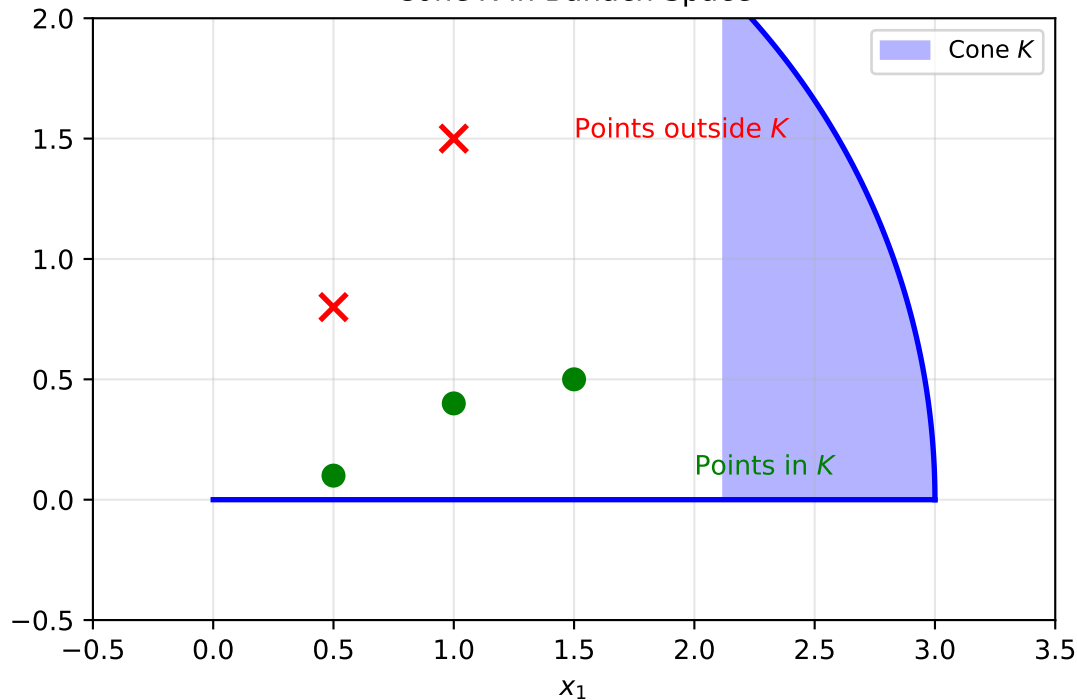


Cone K in Banach Space



Positive Operator Properties

1. Positive: $T(K) \subseteq K$ (maps cone to itself)
2. Strictly Positive: $T(\bar{K}) \subset K$ (interior)
3. Strongly Positive: $T(\bar{K}) \subseteq \text{int}(K)$

Fixed Point Properties:

- Every positive linear operator has a characteristic vector
- Fixed points exist under compactness conditions
- Multiple fixed points possible (order-theoretic methods)